

The Amplified 0/0 Stochastic Resonance and the Rented Repayment

Chapter 70 of the L.O.R.E. Framework

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L.O.R.E. Framework (Law of Repulsive Emanation)

Sigma Chassis v2.0

Abstract

Chapter 69 measured the fluctuation-dissipation pact: the bath repays the demon's bills (Ch.68) as noise at the rate $k_B T$ per degree of freedom. This chapter shows that the repayment is not spent. Faced with a barrier, the noise is re-lent as amplification: a weak, subthreshold signal that cannot cross the barrier alone is carried over it by the very jitter the bath already owes. The amplification is a removable 0/0 in noise intensity D : frozen at $D \rightarrow 0$ (only static compliance, gain $\lim 1$) and buried at $D \rightarrow \infty$ (signal drowned, gain $\lim 0$); it lives between the limits, at an optimal D where the Kramers leak rate of the barrier meets the frequency of the signal. A 2000000-step simulation of the bistable potential $V(x) = x^4/4 - x^2/2$ under subthreshold drive $A = 0.1$ (critical tilt $A^* = 0.3849$) measures the amplification bell: gain 1.16 at $D = 0.03$ to a peak gain 3.36 at $D_{\text{opt}} = 0.155$, sinking toward burial at high D . Kramers' 1940 law is recovered to $DU_{\text{meas}}/DU = 1.031$. At the peak the crossing rate meets the signal rate: $r_{\text{opt}}/f_s = 0.989$. The pattern is confirmed in nature: the 100-kyr ice-age cycle (Benzi et al. 1982), the Schmitt-trigger circuit (Fauve and Heslot 1983), and predatory electroreception (Douglass 1993; Russell 1999).

1. The Double Well and the 0/0 of Amplification

The minimal engine of this chapter is a symmetric bistable well with barrier one quarter above the minima:

$$V(x) = x^4/4 - x^2/2, \quad \text{minima at } x = \pm 1, \quad \text{barrier } DU = 1/4$$

A periodic force below the critical tilt (the tilt at which the barrier vanishes entirely),

$$A^* = 2/(3\sqrt{3}) = 0.3849, \quad \text{drive used: } A = 0.1 \ll A^*$$

cannot alone move the particle from one well to the other. The subthreshold signal merely tilts the floor: a particle held in a well responds with the static compliance $A/|V''| = 0.1/2 = 0.05$, a gain of exactly 1. Against this floor, the story of this chapter is a 0/0:

- $D \rightarrow 0$: no jitter, no escape; the gain is 1, the identity. No amplification.
- $D \rightarrow \infty$: jitter floods the record; the coherent line at the signal frequency vanishes, gain $\rightarrow 0$. No amplification.
- $0 < D < \infty$: the barrier leaks at Kramers' rate, and in the mid-range the leak can be tuned to the signal.

Amplification exists nowhere in the limits and only between them: a removable singularity in noise intensity, exactly the 0/0 that Chapter 1 resolved for $\sin(x)/x$.

2. The Leak: Kramers' Law (1940)

The master crossing frequency of a noisy barrier was derived by Kramers (1940):

$$r(D) = (\sqrt{2}/(2\pi)) * \exp(-DU/D)$$

with prefactor $\sqrt{V''(-1)} * |V(0)|/(2\pi) = \sqrt{2}/(2\pi) = 0.22508$. The simulation counts committed well-to-well transitions (a hysteresis of 0.5 absorbs the top-of-barrier recrossings, the Schmitt-trigger measure). Law-regime rates over 40000 s per noise value:

- $D = 0.070$: $r = 5.925e-3$ vs $6.328e-3$ (ratio 0.936)
- $D = 0.085$: $r = 1.175e-2$ vs $1.189e-2$ (ratio 0.989)
- $D = 0.100$: $r = 1.720e-2$ vs $1.848e-2$ (ratio 0.931)
- $D = 0.115$: $r = 2.573e-2$ vs $2.560e-2$ (ratio 1.005)
- $D = 0.130$: $r = 3.260e-2$ vs $3.290e-2$ (ratio 0.991)

A least-squares Arrhenius fit over the law regime returns $DU_{\text{meas}} = 0.258$ vs 0.250 : $DU_{\text{meas}}/DU = 1.031$. The leak obeys its law.

3. The Bell: Amplification vs Noise

Each noise intensity was run 3 independent trajectories of 3500 s (175 drive cycles each, 2000000 steps total) and the

coherent response at the signal frequency was extracted by phase quadrature. The amplification gain $g = A_{\text{cont}}/(A/|V|^{|f|})$ rises from its identity floor, peaks, and decays:

- $D = 0.03$: $g = 1.16$ = floor (the static tilt, signal no amplification)
- $D = 0.09$: $g = 2.01$
- $D = 0.13$: $g = 3.06$
- $D = 0.155$: $g = 3.36$ <- measured peak (236% over the floor)
- $D = 0.22$: $g = 3.30$
- $D = 0.70$: $g = 2.47$ (decaying)
- $D = 1.50$: $g = 1.84$; $D = 2.00$: $g = 2.08$ (sinking toward burial)

The two-state (sign x) telegraph gives the classic SNR bell: 0.0000 at $D = 0.03$, rising to 0.0143 at $D = 0.22$, falling to 0.0025 at $D = 1.50$. A signal that the barrier alone cannot pass is amplified nearly three and a half times by the noise that Ch.69 measured as pure dissipation.

4. The Synchrony: Where the Leak Meets the Beat

The peak is not arbitrary. At the measured optimum $D_{\text{opt}} = 0.155$ the barrier leak rate coincides with the drive frequency:

$$r_{\text{opt}}/f_s = 0.0494/0.050 = 0.989 \quad (\text{theory } r(D_{\text{opt}})/f_s = 0.898)$$

The matching point $r(D) = f_s$ sits at $D^* = 0.166$, against $D_{\text{opt}} = 0.155$: $D_{\text{opt}}/D^* = 0.933$. The removable value of the amplification 0/0 is the noise at which the bath leaks the barrier once per signal period: gain is maximized exactly when the well is bled in time with the signal. The repayment is no longer spent; it is lent at the frequency of the borrower.

5. Nature Rents the Noise

The pattern was first found in the ice ages. Milankovitch (1941) showed the insolation forcing; Benzi, Parisi, Sutera and Vulpiani (1982) showed that its weak 100-kyr component - too weak to push the climate across its glacial/inter glacial barrier alone - is amplified by the noise of the climate system itself. Vostok ice records (Petit et al. 1999) document the 100-kyr cycle with a subthreshold Milankovitch drive: stochastic resonance in a bistable climate.

The laboratory and the organism agree:

- Fauve and Heslot (1983): first laboratory measurement of the SR bell in a Schmitt trigger - the exact committed-well device of this chapter.
- Douglass, Wilkens, Pantazelou and Moss (1993): noise enhances signal detection in crayfish mechanoreceptors; a peak in the SNR-vs-noise curve at the optimal noise.
- Longtin, Bulsara and Moss (1991): stochastic resonance in integrate-and-fire neurons.
- Russell, Wilkens and Moss (1999): the paddlefish detects prey by amplifying a subthreshold electrosensory signal with the noise it already has.
- Gammaitoni, Hanggi, Jung and Marchesoni (1998): the review consolidating SR across physics, electronics, chemistry and biology - a genuinely nonlinear effect.

In every case the organism or system does not buy noise; it rents the noise the bath already pays. The paddlefish owes nothing for its gain: the repayment of Ch.69 is simply lent across its prey-detection barrier.

6. Theorem and Resolution

Theorem (The Amplified 0/0). Let V be bistable with barrier DU and noise intensity D , driven subthreshold at frequency f_s . The coherent gain $g(D)$ satisfies:

- $g(0) = 1$ (static compliance; zero amplification)
- $g(\infty) = 0$ (buried; zero amplification)

- g attains a maximum $g^* > 1$ at D_{opt} where the Kramers rate $r(D_{\text{opt}}) \sim f_s$.

Resolution. Amplification is 0/0 in the noise: absent at no noise, absent at all noise, removable between. The removable value is set by the synchrony of leak and beat. No free lunch: the gain is the Ch.69 repayment re-lent, and nothing is created. The ledger of the bath closes: the demon bills (Ch.68), the bath repays (Ch.69), and the barrier borrows back (Ch.70) - at interest measured here as g^* minus 1, 236%, payable not by the signal but by the noise already owed.

7. Conclusion

Chapter 70 converts the fluctuation-dissipation pact of Chapter 69 from a cost into a bank. Kramers' leak ($DU_{\text{meas}}/DU = 1.031$), the amplification bell (g 3.36 at D_{opt} 0.155), and the synchrony identity ($r_{\text{opt}}/f_s = 0.989$, $D_{\text{opt}}/D^* = 0.933$) are measured to the decadal percent, and the same numbers are found in the ice ages, the Schmitt trigger, the neuron, and the paddlefish. The noise is not the payment; the noise is the loan.

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